

Enhancing Mathematics Achievement in Elementary Education

Evidence from a Cluster-Randomized Controlled Trial

Intervention Instrument:

Mia's Macondo Adventure

(Multi-User Gamified AI Math Platform)

Submitted for Review · June 2026

Keywords: human capital, ed-tech, randomized controlled trial,
artificial intelligence, dashboards, primary education

JEL Codes: I21, I28, C93, O33, D83

Abstract

Foundational mathematics skills acquired during the final years of elementary education are critical predictors of long-term human capital formation. However, structural constraints, including heterogeneous classroom proficiency levels and limited real-time diagnostic data for teachers and policymakers, persistently hinder learning gains.

This proposal details the design of a cluster-randomized controlled trial (RCT) to evaluate the causal impact of *Mia’s Macondo Adventure*—a gamified, AI-powered mathematics platform—on standardized math scores among final-year elementary students. The intervention features a unique multi-user architecture: interactive AI-driven learning modules for students, progress-monitoring dashboards for teachers, aggregation tools for school directors, and macro-level performance heatmaps for local government officials.

We propose a school-level cluster-randomized design across 80 schools (40 treatment, 40 control), stratified by municipality, school size, and historical academic performance. The primary estimation strategy relies on Ordinary Least Squares (OLS) with baseline controls to maximize precision. In addition to primary intent-to-treat (ITT) estimates on standardized test scores, we specify a Local Average Treatment Effect (LATE) framework to measure the impact of platform compliance, and explore heterogeneous treatment effects to understand how AI design interacts with baseline proficiency.

Contributions: (i) robust experimental evidence on the effectiveness of interactive AI in reducing educational disparities; (ii) a causal evaluation of how multi-tier data visibility (from classroom to municipality) influences pedagogical and administrative decision-making; and (iii) a scalable framework for integrating ed-tech within existing public education infrastructure.

Contents

1	Introduction	2
1.1	Research Questions	2
2	Intervention: <i>Mia’s Macondo Adventure</i>	3
2.1	Student Interface: Gamification and AI Tutoring	3
2.2	Multi-Tiered Dashboard Architecture	3
3	Empirical Strategy	3
3.1	Unit of Randomization	4
3.2	Sampling and Stratification	4
3.3	Power Calculation for Cluster RCT	4
3.4	Estimation: Ordinary Least Squares (OLS)	5
3.5	Compliance and Instrumental Variables (LATE)	5
4	Implementation and Data Collection	5
4.1	Outcome Measurement	5
4.2	Threats to Validity & Mitigations	6
5	Timeline and Pre-Registration	6

1. Introduction

Human capital accumulation fundamentally relies on the mastery of foundational mathematics during primary education. Deficits at this stage compound over time, leading to lower secondary school completion rates, reduced access to higher education, and constrained labor market outcomes. While traditional interventions frequently target input-based constraints (e.g., textbook provision or class size reduction), the economics of education literature increasingly highlights that learning gaps are driven by a mismatch between the rigid pace of standardized curricula and the heterogeneous learning levels of individual students.

Technological interventions offer a theoretical mechanism to resolve this mismatch. Adaptive, AI-driven educational technology (ed-tech) can tailor instruction to a student's frontier of knowledge, providing the targeted feedback characteristic of one-on-one tutoring at a fraction of the cost. However, the existing experimental evidence on ed-tech is highly mixed. Interventions that simply digitize standard textbooks or lack integration with the broader pedagogical ecosystem frequently yield null effects. Success largely depends on two factors: the sustained engagement of the student and the actionable integration of the technology into the teacher's workflow and the local government's resource allocation strategy.

Mia's Macondo Adventure is engineered specifically to address these institutional and behavioral constraints. It is a gamified, multi-user mathematics platform. For the student, it utilizes conversational AI chatbots to dynamically influence engagement and scaffold learning in topics such as logic, fractions, and geometry. Crucially, the platform simultaneously provisions real-time diagnostic dashboards tailored to three higher-level stakeholders: teachers, school directives, and local educational ministries.

This proposal outlines the design of a cluster-randomized controlled trial (RCT) to estimate the causal impact of this platform. The research design mitigates spillovers through school-level randomization and maximizes statistical power using baseline-adjusted OLS specifications.

1.1 Research Questions

1. Does school-level access to the *Mia's Macondo Adventure* platform improve standardized math scores among final-year elementary students compared to business-as-usual instruction?
2. Conditional on access, what is the Local Average Treatment Effect (LATE) of active platform engagement (compliance) on student learning gains?
3. How do multi-tier administrative dashboards influence teacher behavior, school-level resource allocation, and local government decision-making?
4. Are the learning effects heterogeneous across baseline proficiency levels, school resource levels, or urban/rural divides?

2. Intervention: *Mia's Macondo Adventure*

The intervention moves beyond isolated student software by implementing a systemic, multi-user architecture designed to bridge the gap between individual learning and institutional policy.

2.1 Student Interface: Gamification and AI Tutoring

The core learning engine is driven by a narrative gamification structure. Students navigate through modules mapped to the national curriculum (e.g., Logic, Ratios, Fractions, Geometry). The platform employs AI chatbots that mimic a tutor’s behavior—eliciting reasoning from the student rather than simply providing the correct answer. This “tailoring premium” keeps students in their zone of proximal development. Success in exercises yields Experience Points (XP), unlocking new narrative levels and maintaining sustained engagement.

2.2 Multi-Tiered Dashboard Architecture

The defining feature of this intervention is the routing of real-time data to decision-makers, tailored to their operational scope:

Table 1: Dashboard Data Access by Stakeholder

User Role	Granularity	Key Metrics & Actionable Data
Student	Individual	XP accumulated, modules unlocked, personalized error correction.
Teacher	Classroom	Module completion rates, alerts for inactive students (no login in 7+ days), identification of specific concepts causing class-wide bottlenecks.
Directive	School	Aggregate completion across classes, inter-classroom variance, attendance proxy (active days/week).
Local Gov.	Municipality	School-level aggregate performance, urban vs. rural proficiency gaps, real-time macro diagnostic of curriculum progression.

Providing real-time treatment/control outcome comparisons to local government officials during an ongoing RCT generates extreme political pressure to intervene, unblind, or expand the treatment prematurely, contaminating the research design. Therefore, the **Local Government Dashboard will be strictly read-only for outcome data during the active trial phase**. Officials will see aggregate engagement data but will not see performance comparisons between treatment and control schools until the study concludes.

3. Empirical Strategy

3.1 Unit of Randomization

We utilize a **school-level cluster-randomized design**. Randomizing at the student or classroom level in ed-tech evaluations frequently introduces severe biases:

- **SUTVA Violations (Peer Spillovers):** Treated students share devices or accounts with control students.
- **Teacher Spillovers:** A teacher adjusts their pedagogical pace for the entire classroom based on the supplemental learning of treated students, contaminating the control group.

Randomizing at the school level establishes clean control units and accurately reflects the policy reality of how software licenses are procured and deployed by local governments.

3.2 Sampling and Stratification

The target population comprises students in their final year of elementary education (typically grade 5 or 6). To ensure balance and optimize precision, we perform a stratified block randomization prior to treatment assignment. Strata are constructed using:

- Municipality / Geographic region
- School size (terciles)
- Historical baseline math performance
- Urban / Rural classification

Within each stratum, schools are assigned 1:1 to either the treatment group (platform access) or the control group (business-as-usual).

3.3 Power Calculation for Cluster RCT

Because students within the same school share unobserved characteristics and shocks, errors are correlated. The power calculation must account for the Intraclass Correlation Coefficient (ρ_{icc}) via the Design Effect ($DEFF = 1 + (m - 1)\rho_{icc}$), where m is the number of sampled students per school.

- **Target Minimum Detectable Effect (MDE):** 0.20 Standard Deviations (SD).
- **Parameters:** Power $(1 - \kappa) = 0.80$, Significance $(\alpha) = 0.05$.
- **Cluster specifics:** Average cluster size $m \approx 35$ students. Assumed $\rho_{icc} = 0.15$.
- **Variance explained by baseline covariates (R^2):** 0.40 (incorporating baseline test scores substantially reduces residual variance).

Under these parameters, achieving 80% power requires approximately **80 schools (40 treatment, 40 control)**, yielding a total sample of roughly 2,800 students.

3.4 Estimation: Ordinary Least Squares (OLS)

The preferred estimator in applied microeconomics for this design is Ordinary Least Squares (OLS) controlling for baseline outcomes. The primary Intent-to-Treat (ITT) specification is:

$$Y_{ijst} = \alpha + \beta \cdot \text{Treat}_s + \gamma \cdot Y_{ijst-1} + \mathbf{X}'_{ijst}\delta + \mu_{\text{stratum}} + \varepsilon_{ijst} \quad (1)$$

Where:

- Y_{ijst} is the standardized endline math score for student i in classroom j , school s , at time t .
- Treat_s is a binary indicator equalling 1 if school s was randomized to treatment.
- Y_{ijst-1} is the baseline math score (the critical control variable for precision).
- $\mathbf{X}_{ijst}$ is a vector of student and teacher covariates (gender, SES proxy, teacher experience).
- μ_{stratum} are randomization strata fixed effects.
- ε_{ijst} is the idiosyncratic error term, **clustered at the school level** to account for intra-cluster correlation.

The parameter of interest is β , which captures the ITT policy effect: the impact of providing a school access to the platform, regardless of individual student usage.

3.5 Compliance and Instrumental Variables (LATE)

Ed-tech deployments routinely suffer from partial compliance; some assigned schools may fail to integrate the platform, and specific students may rarely log in. To understand the effect of actual usage, we define compliance via platform telemetry (e.g., completing at least 3 curriculum modules). We estimate the Local Average Treatment Effect (LATE) via Two-Stage Least Squares (2SLS), instrumenting actual engagement (E_{ijst}) with random assignment (Treat_s):

$$E_{ijst} = \pi_0 + \pi_1 \cdot \text{Treat}_s + \pi_2 \cdot Y_{ijst-1} + \mathbf{X}'_{ijst}\phi + \mu_{\text{stratum}} + \nu_{ijst} \quad (2)$$

$$Y_{ijst} = \alpha + \beta_{\text{LATE}} \cdot \hat{E}_{ijst} + \gamma \cdot Y_{ijst-1} + \mathbf{X}'_{ijst}\delta + \mu_{\text{stratum}} + \varepsilon_{ijst} \quad (3)$$

4. Implementation and Data Collection

4.1 Outcome Measurement

To prevent “teaching to the test,” the primary outcome must be measured via an independent, external standardized evaluation administered by the research team, explicitly mapped to the national curriculum logic, fractions, and geometry standards.

Table 2: Variables and Data Sources

Type	Variable	Source
Primary	Standardized math test score (endline, z -scored)	Independent evaluation
Secondary	In-app correct answer rates and XP	Platform server logs
Secondary	Time-on-task (active minutes per week)	Platform server logs
Secondary	Teacher & Directive dashboard engagement	Platform server logs
Mechanism	Frequency of teacher pedagogical interventions	Teacher endline survey

4.2 Threats to Validity & Mitigations

Hawthorne Effect: Control schools might exert extra effort simply because they are being observed. Mitigation: Administer baseline and endline testing as part of standard municipal evaluations, masking the specific digital intervention focus from control schools where possible.

Differential Attrition: Students in treatment schools might be more likely to attend on endline test day. Mitigation: We will bound effects using Lee (2009) bounds if differential attrition across arms is detected.

5. Timeline and Pre-Registration

Before any data collection occurs, the trial will be pre-registered on the AEA RCT Registry. The Pre-Analysis Plan (PAP) will explicitly define the OLS specification, covariate selection, and index construction to prevent p-hacking.

Table 3: Implementation Timeline

Phase	Activity	Duration
T-6 months	Baseline standardized math testing across all 80 sample schools	2 weeks
T-5 months	Stratified randomization and stakeholder notification	1 week
T-4 months	Platform deployment, teacher/directive training in treatment schools	3 weeks
T=0 to T+6	Active intervention period (one full academic semester)	6 months
T+6 months	Endline standardized math test (independent administration)	2 weeks
T+8 months	Data cleaning, OLS estimation, dashboard unlock for local gov.	2 months